



Using Bayer Pattern to Detect Fake Images

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Fake Images

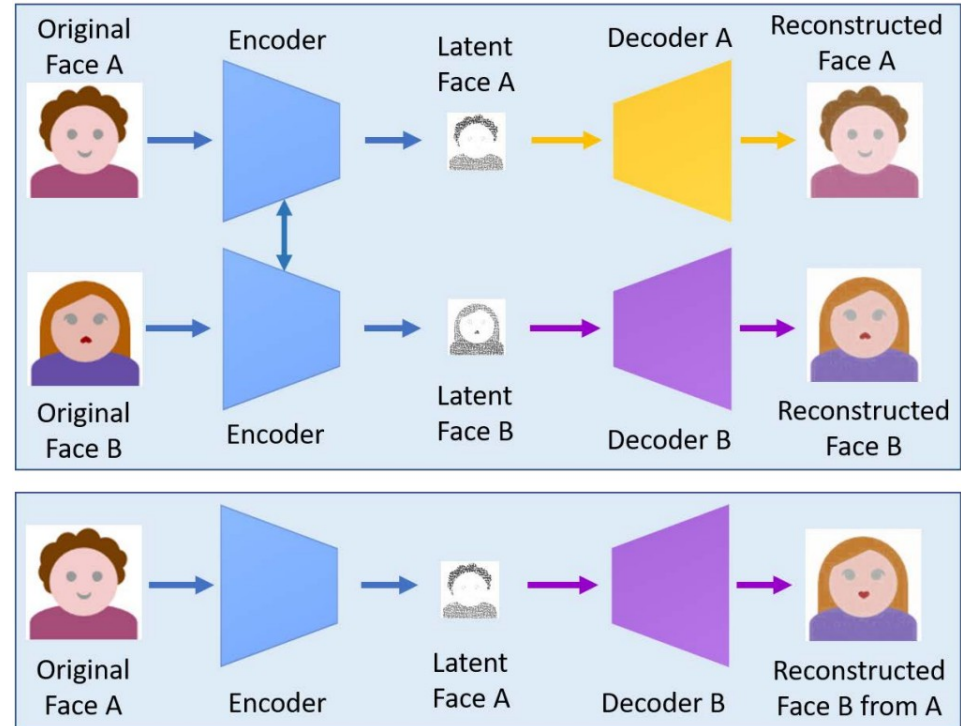
Computer synthesised images

- Variational autoencoder (VAE)
- Generative adversarial network (GAN)
- Diffusion model



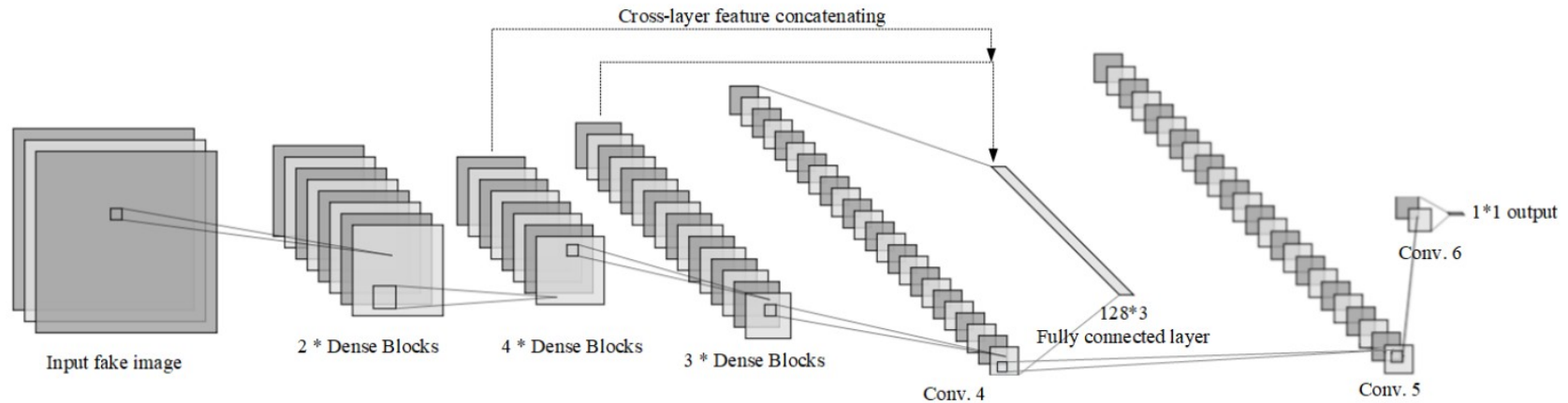
Deepfake

Replace one identity's face with that of another while maintaining the original facial expressions



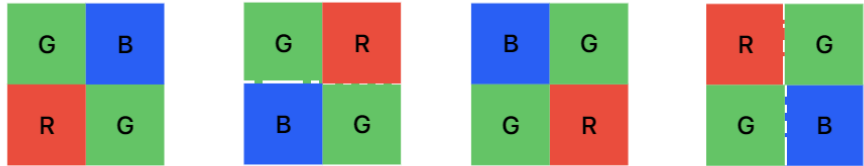
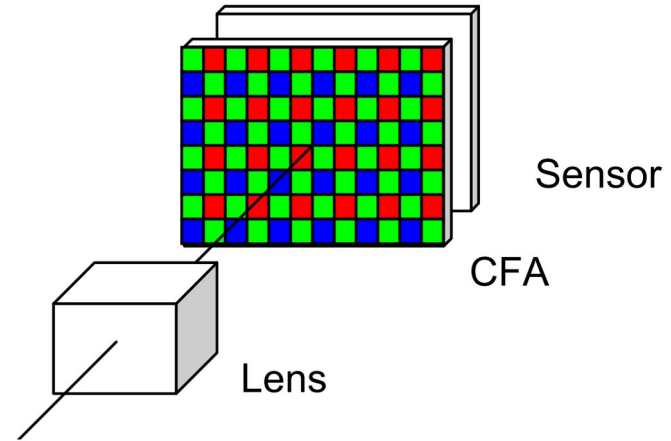
Nguyen, Thanh Thi, et al. "Deep learning for deepfakes creation and detection: A survey." *Computer Vision and Image Understanding* 223 (2022): 103525.
<https://www.theguardian.com/technology/ng-interactive/2019/jun/22/the-rise-of-the-deepfake-and-the-threat-to-democracy>

Previous works - Deep Learning

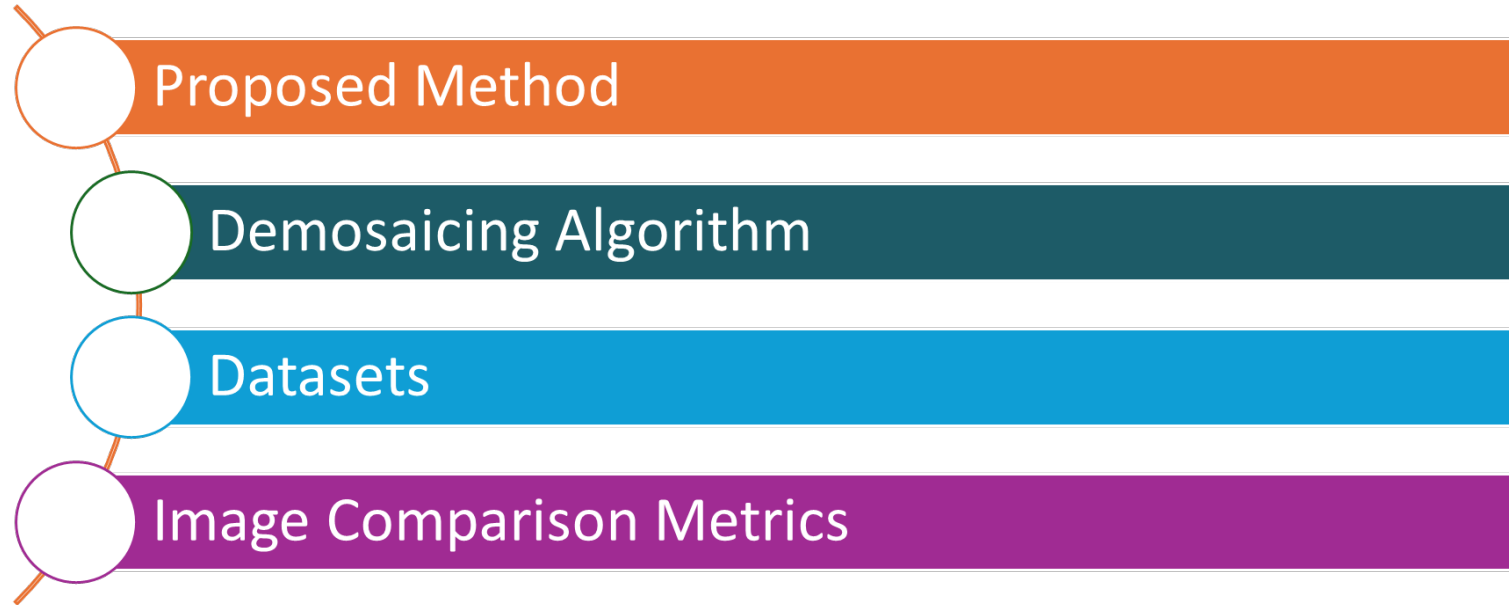


Bayer pattern

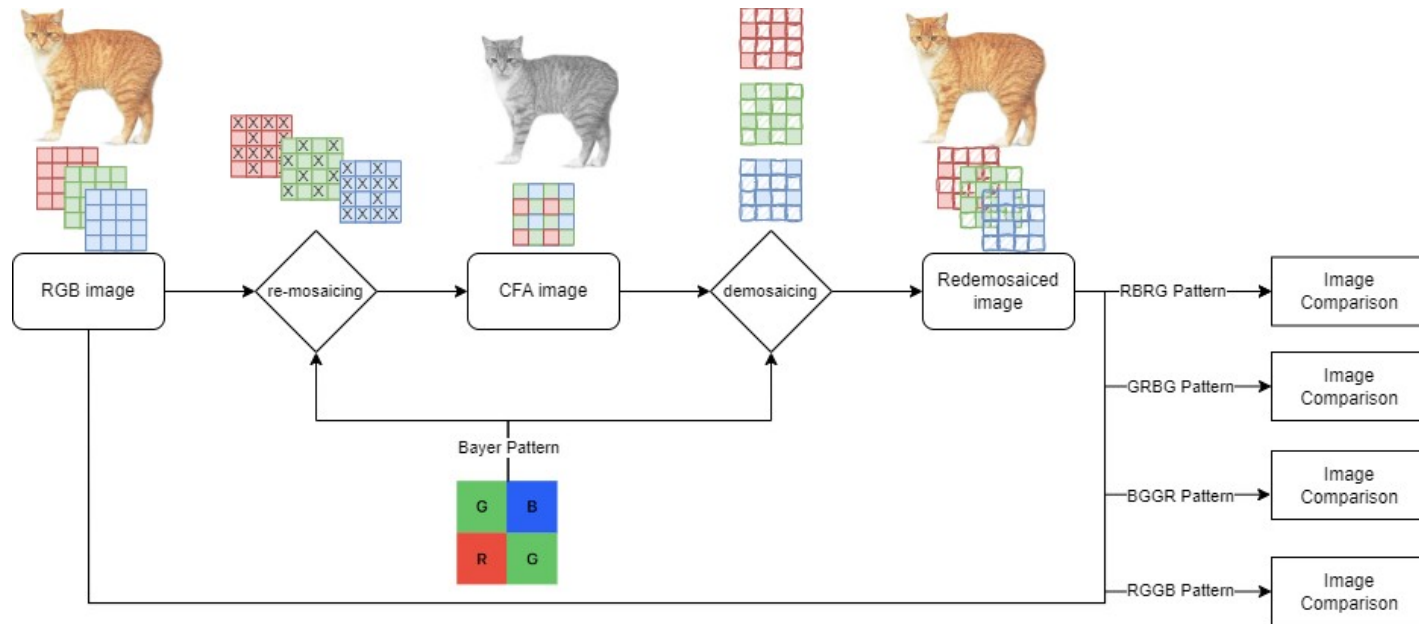
- Color filter array(CFA): Bayer pattern
- 4 Bayer patterns:
GBRG, GRBG, BGGR, RGGB
- Demosaicing: reconstruct missing color for all RGB channels



Methodology

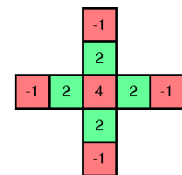


Proposed Method

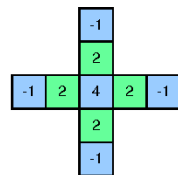


Demosaicing Algorithm

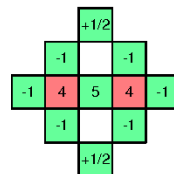
- Developed by Malvar et al in 2004
- Gradient-corrected form of bilinear interpolation
- Implemented in our project using an open-source Python version of this algorithm.
- The code was modified and fine tuned to match the results from the MATLAB implementation of this method.



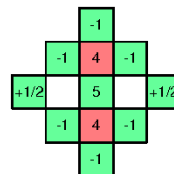
G at R locations



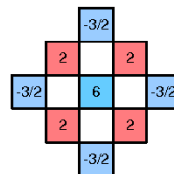
G at B locations



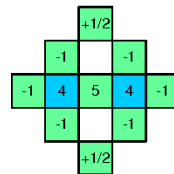
R at green in
R row, B column



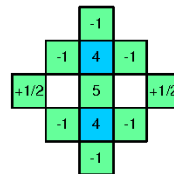
R at green in
B row, B column



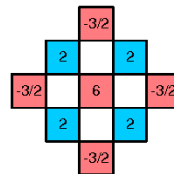
R at blue in
B row, B column



B at green in
B row, R column



B at green in
R row, B column



B at red in
R row, R column

Datasets

We used 4 different datasets for our project which are:



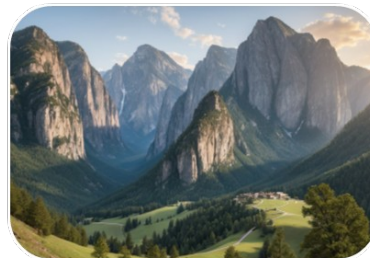
RAISE Dataset



DiffusionDB
Dataset



GenImage Dataset



Landscape
Photoreal Dataset

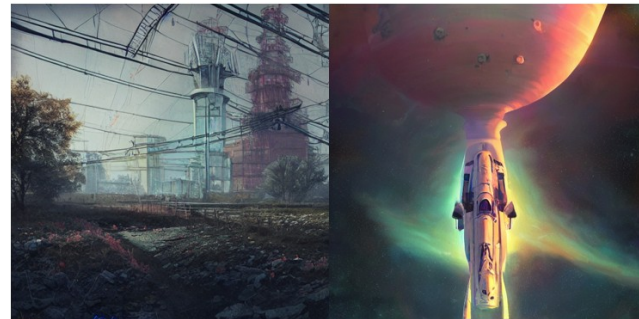
Datasets: RAISE Dataset [Real Images]

- Contains 8,156 RAW high-resolution, camera-native images
- The images were captured using 3 camera models:
 - Nikon (D40 || D90 || D7000)
- The images in this set are grouped into 7 classes.
 - Outdoor, Indoor, Landscape, Nature, People, Objects and Buildings.
- We used the RAISE 1k subset of this dataset, which contains 1000 images, for our experiment.



Datasets: DiffusionDB Dataset [Fake Images]

- Contains 14 million high-resolution, user-generated images.
- Images in this dataset were created using Stable Diffusion.
- The images were curated and compiled by scraping through the official Stable Diffusion Discord page
- We used 1000 images from this dataset in our experiments.



Datasets: GenImage Dataset [Fake Images]

- This dataset contains about 2.68 million images.
- The images are split into real or fake images, which can be paired.
- Fake images were generated using 8 different AI models.
- This dataset is modeled after ImageNet, so it has the same 1000 classes.



Datasets: Landscape PhotoReal Dataset [Fake Images]

- For further experimentation, we created this dataset
- The images were generated using the Landscape PhotoReal generative model. It is a fine-tuned stable-diffusion v1.5 model.
- We created 1000 images with resolution (1024×768)
- Our aim was to create artificial images that look realistic and are not easily detected visually.

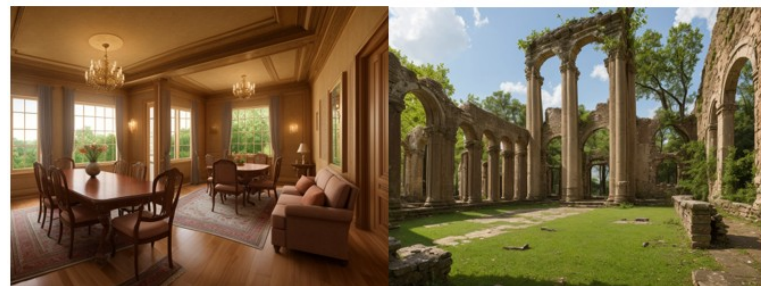
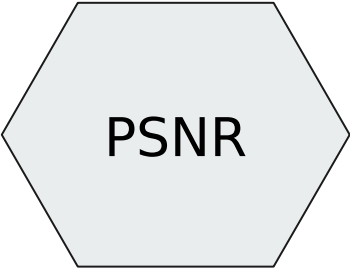
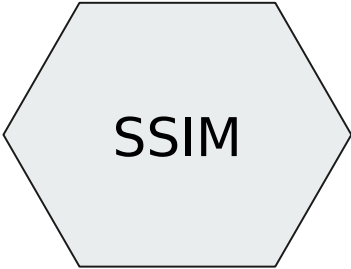


Image Comparison Metrics

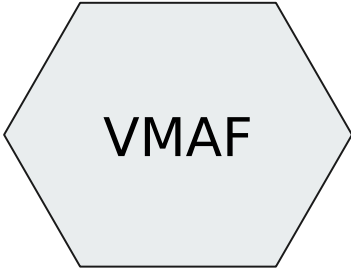
The four metrics used in the evaluation of our proposed method are shown below



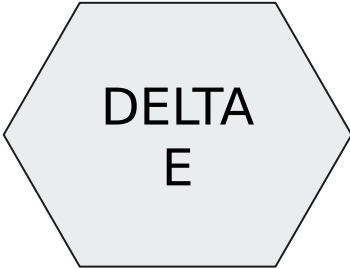
PSNR



SSIM



VMAF



DELTA
E

Image Comparison Metrics

PSNR

- Ratio of peak signal power to the power of noise that distorts the signal.
- Measured in dB
- The higher the PSNR values, the better the quality
- Limitation: It does not correlate

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

SSIM

- This metric aligns better with the HVS
- Image quality is quantified using structure, luminance and contrast.
- Values range between 0 and 1
- Limitation: Performs poorly on low quality images.

$$SSIM(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}.$$

Image Comparison Metrics

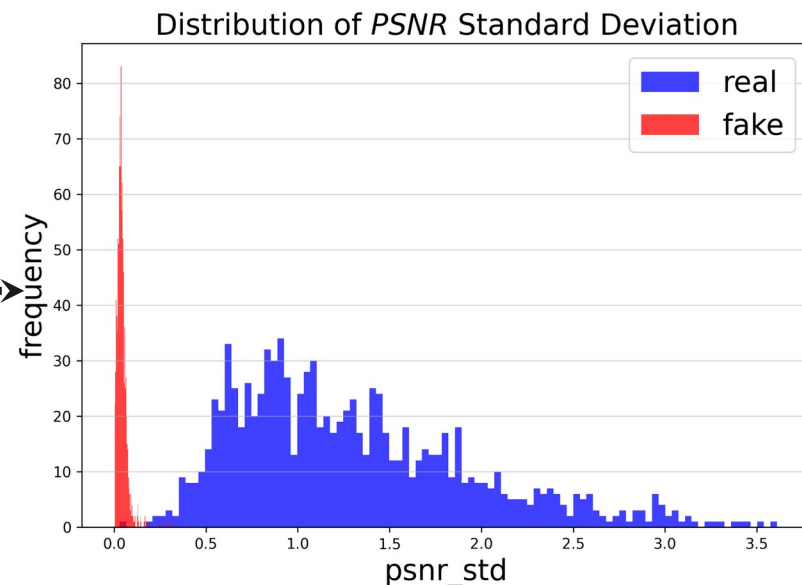
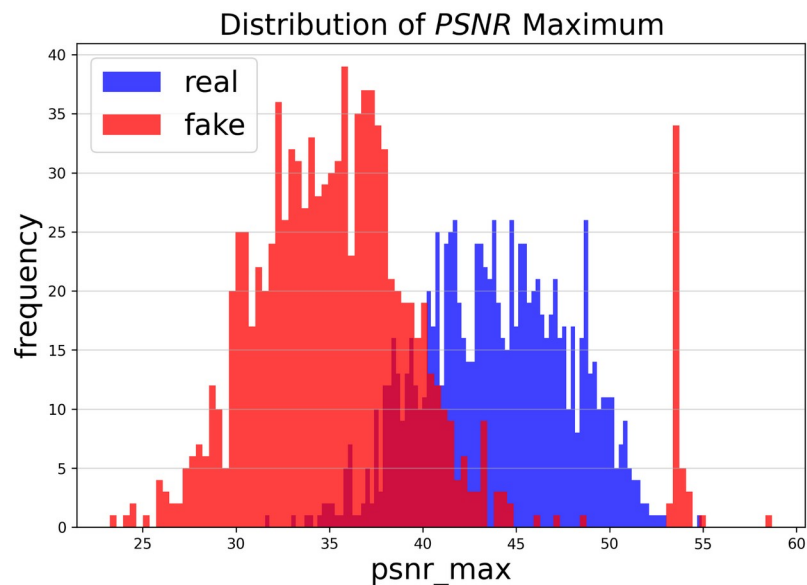
VMAF

- Video quality metric created by Netflix
- Creates a score between 0-100
- Aligns closely with results from subjective tests and human perception.
- We used both versions of VMAF for our experiments
 - VMAF 1080p
 - VMAF 4K

DELTA E

- Created by International Commission on Illumination (CIE)
- Measures the distance between colours in the CIELAB colour space.
- Colour accuracy is the key focus on this metric
- We used the CIE 2000 version of Delta E for our experiments.

Experiments: from Extrema to Standard Deviation

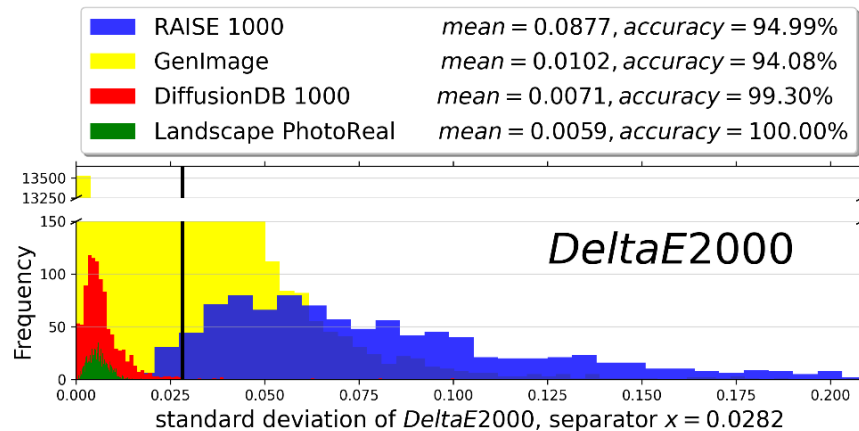
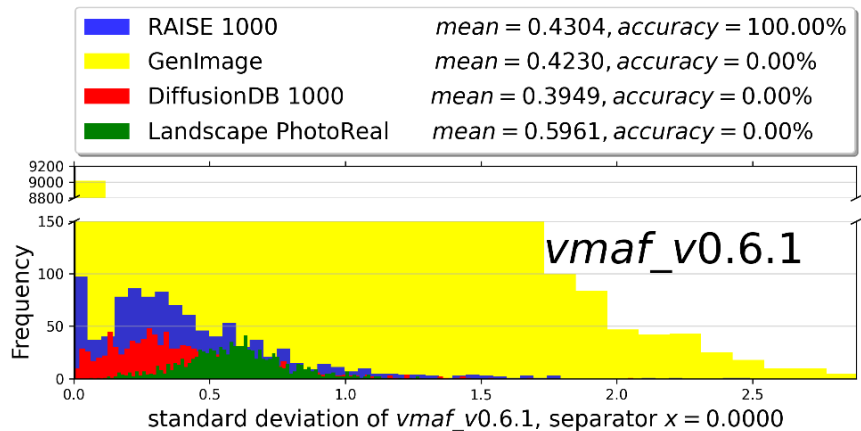
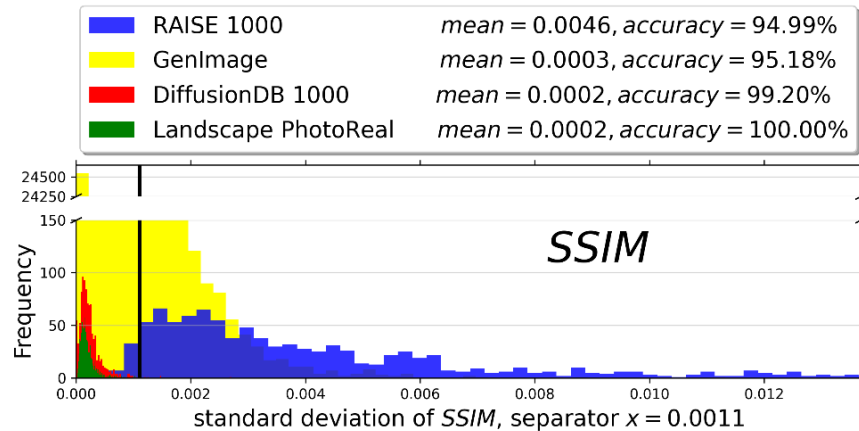
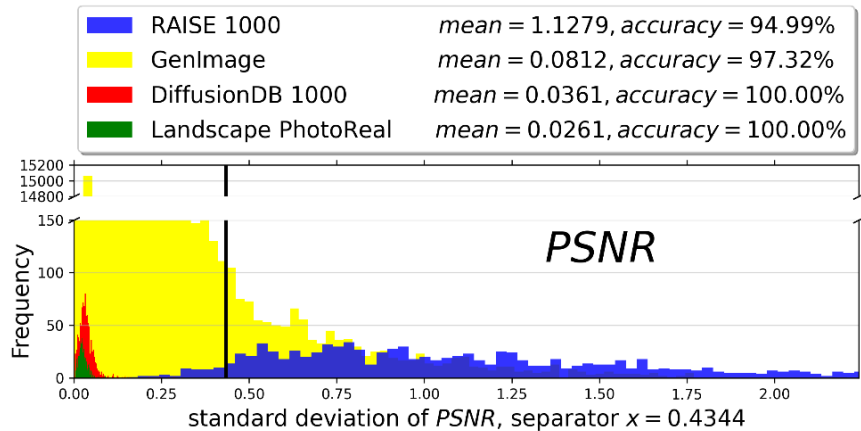


Real vs. Fake: PSNR of Bayer Patterns

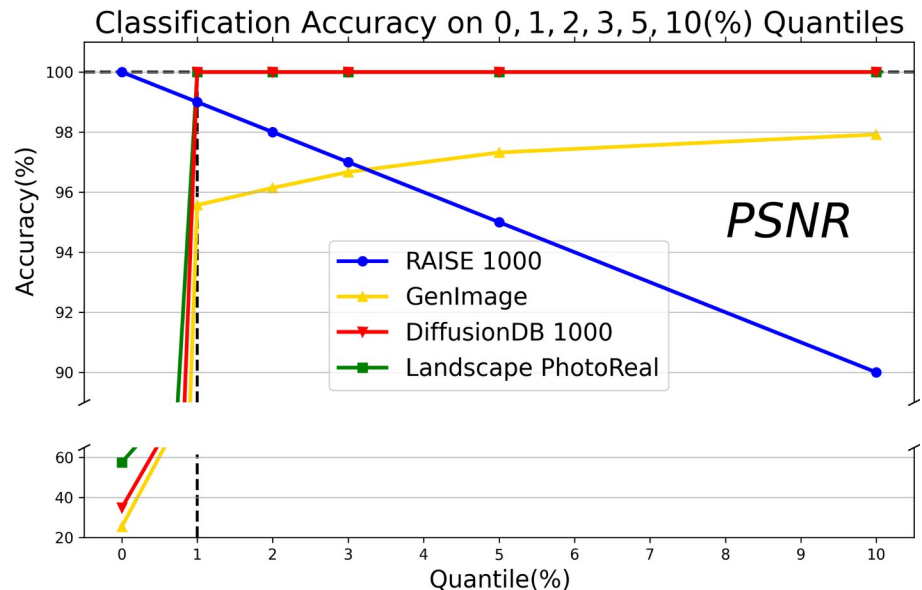
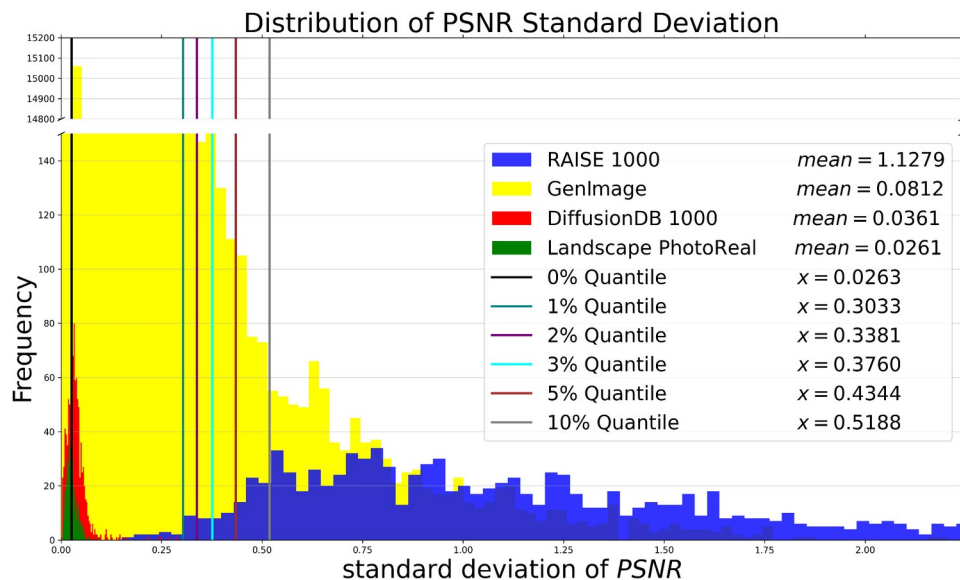
```
"real.TIF": {  
  "true_pattern": "rggb",  
  "psnr": {  
    "gbrg": 39.582973,  
    "grbg": 39.588863,  
    "bggr": 40.296097,  
    "rggb": 40.673027  
  },  
  "psnr_std": 0.468673  
}
```

```
"fake.png": {  
  "true_pattern": "None",  
  "psnr": {  
    "gbrg": 39.388031,  
    "grbg": 39.459885,  
    "bggr": 39.382214,  
    "rggb": 39.459785  
  },  
  "psnr_std": 0.037413  
}
```

Results: 4 Metrics on 4 Datasets, using 5% Quantile



Results: PSNR on 4 Datasets, using 0,1,2,3,5,10(%) Quantiles



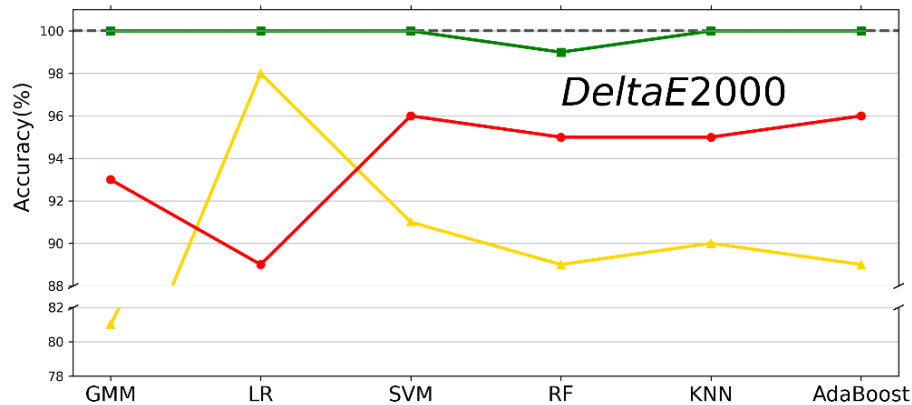
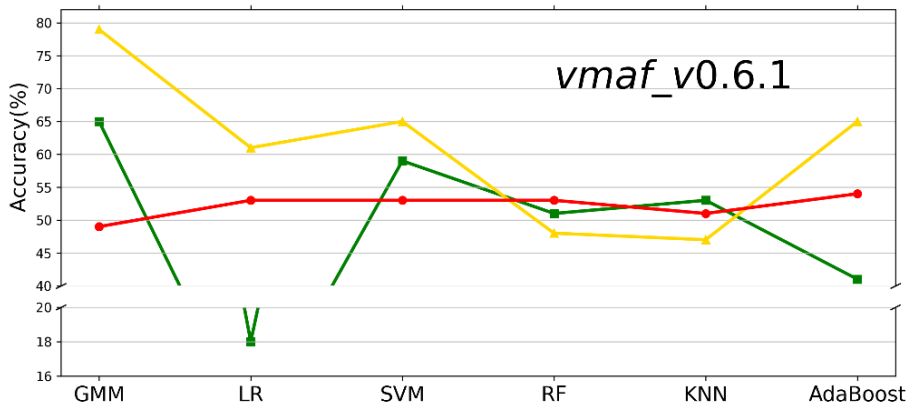
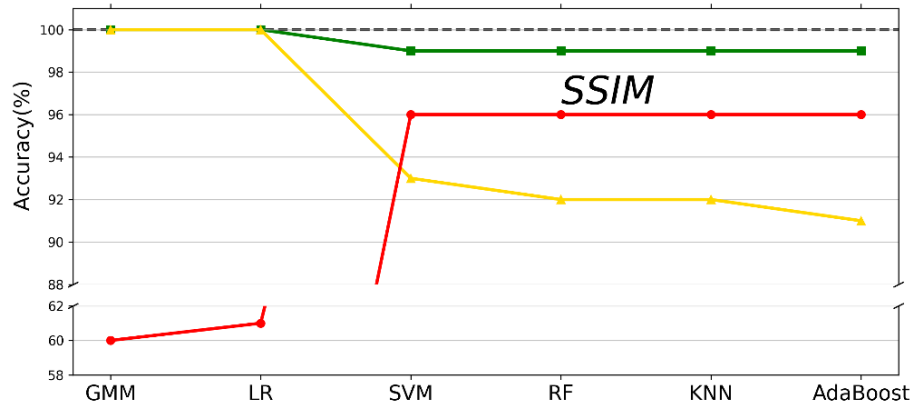
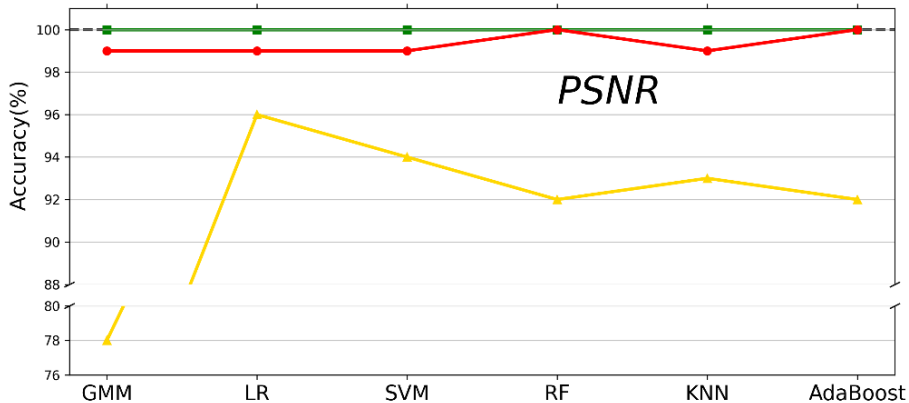
PSNR \ Dataset s	RAISE 1000	GenImage	DiffusionDB 1000	Landscape PhotoReal
1% Quantile	99%	95.57%	100%	100%

Results: 4 Metrics on 3 Test Sets, using 6 ML Algorithms

20%(RAISE+DiffusionDB)

GenImage

Landscape PhotoReal

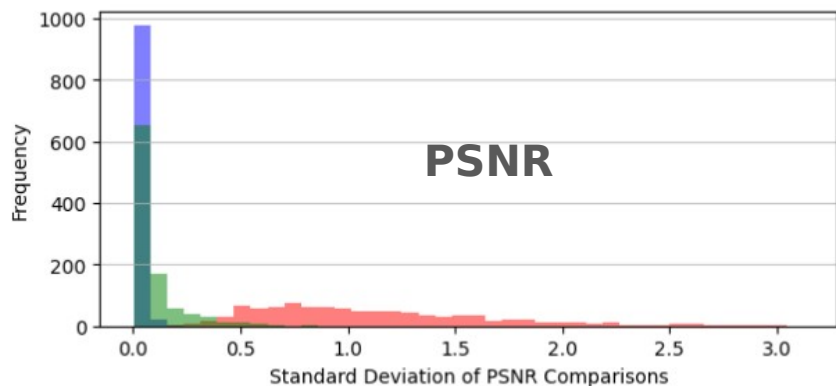




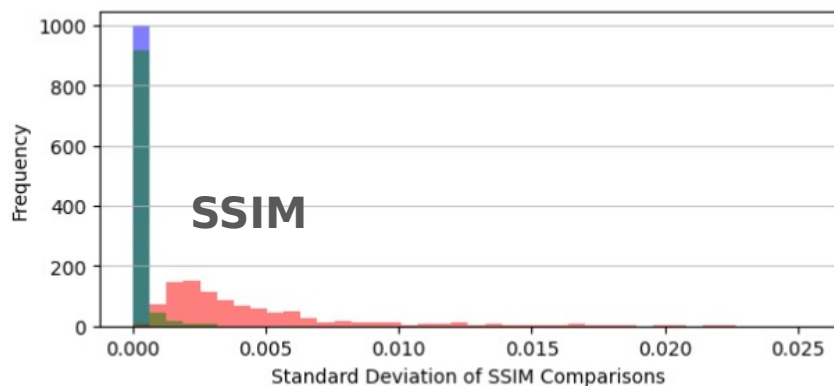
JPEG Compressed Real Images as Reference

- Instead of using original images in the RAISE 1000 dataset
- Use JPEG compression to compress the images
 - Quality 25
 - Quality 75

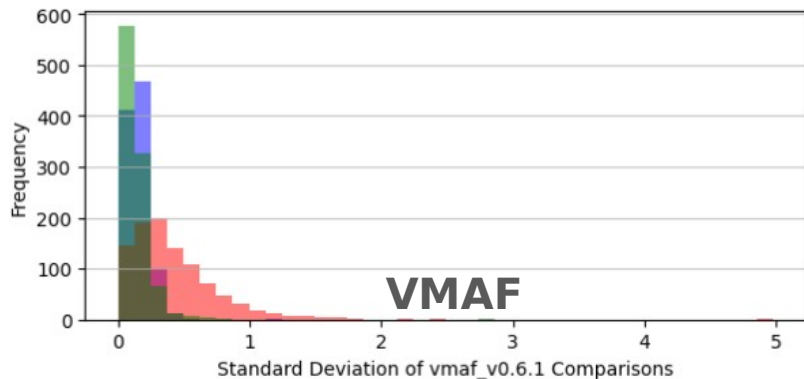
Quality 25 Compressed RAISE 1000 PSNR, mean=0.0280, min=0.0009, max=0.2909
 RAISE 1000 PSNR, mean=1.1279, min=0.0263, max=3.1257
 Quality 75 Compressed RAISE 1000 PSNR, mean=0.1053, min=0.0022, max=1.6278



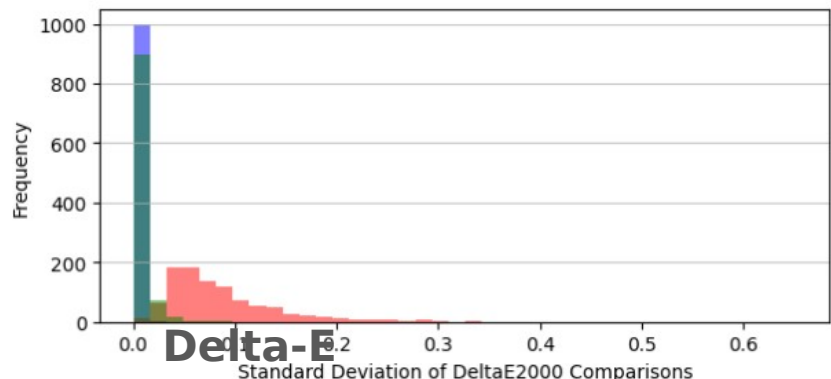
Quality 25 Compressed RAISE 1000 SSIM, mean=0.0000, min=0.0000, max=0.0006
 RAISE 1000 SSIM, mean=0.0046, min=0.0001, max=0.0252
 Quality 75 Compressed RAISE 1000 SSIM, mean=0.0002, min=0.0000, max=0.0155

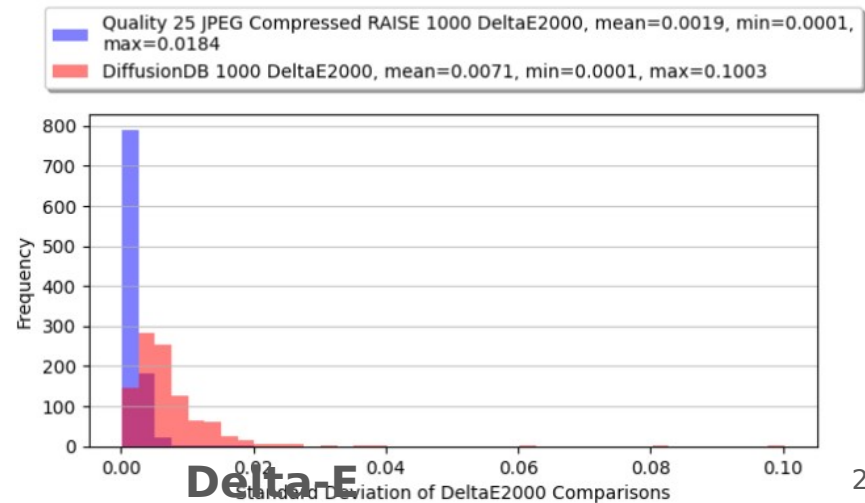
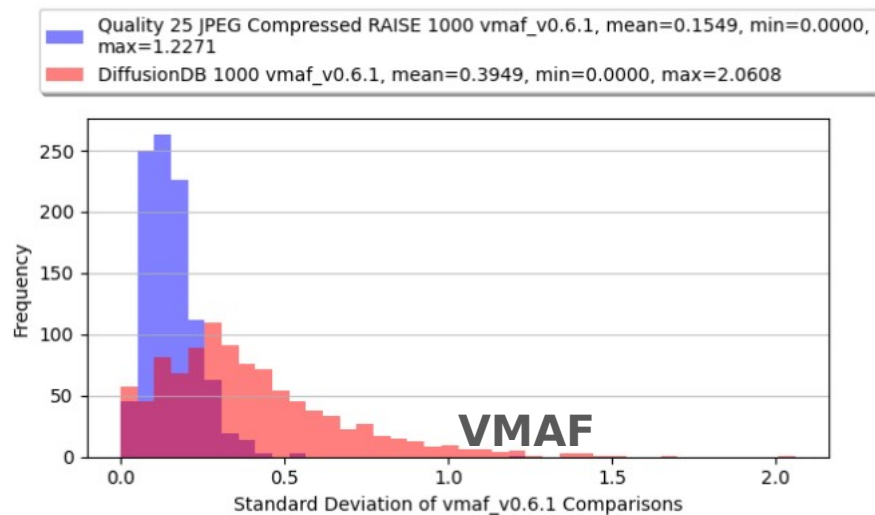
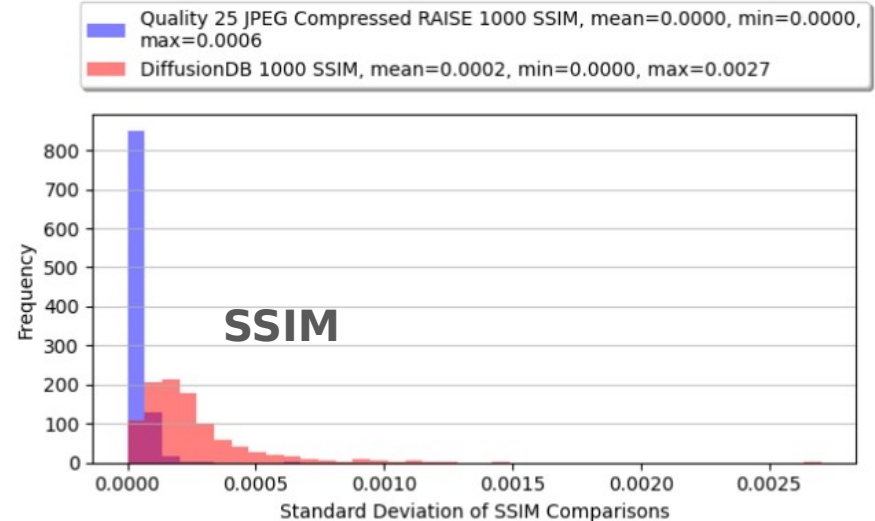
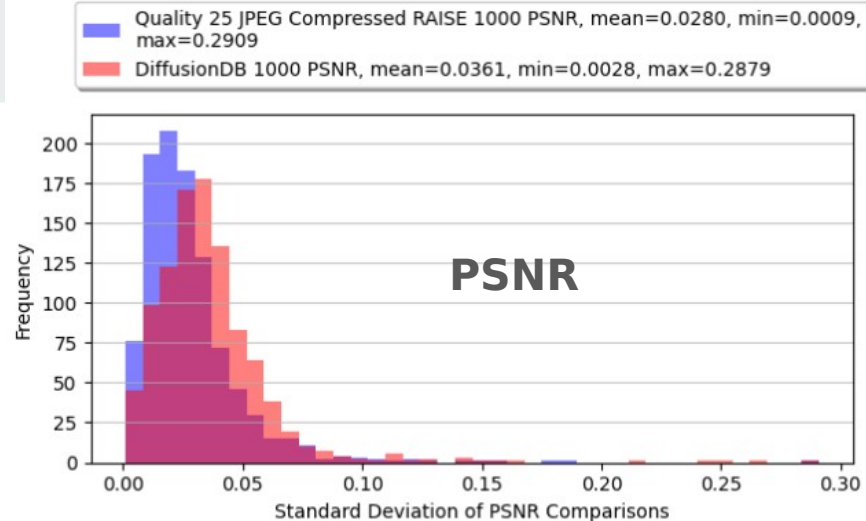


Quality 25 Compressed RAISE 1000 vmaf_v0.6.1, mean=0.1549, min=0.0000, max=1.2271
 RAISE 1000 vmaf_v0.6.1, mean=0.4304, min=0.0000, max=4.9799
 Quality 75 Compressed RAISE 1000 vmaf_v0.6.1, mean=0.1323, min=0.0000, max=2.7424

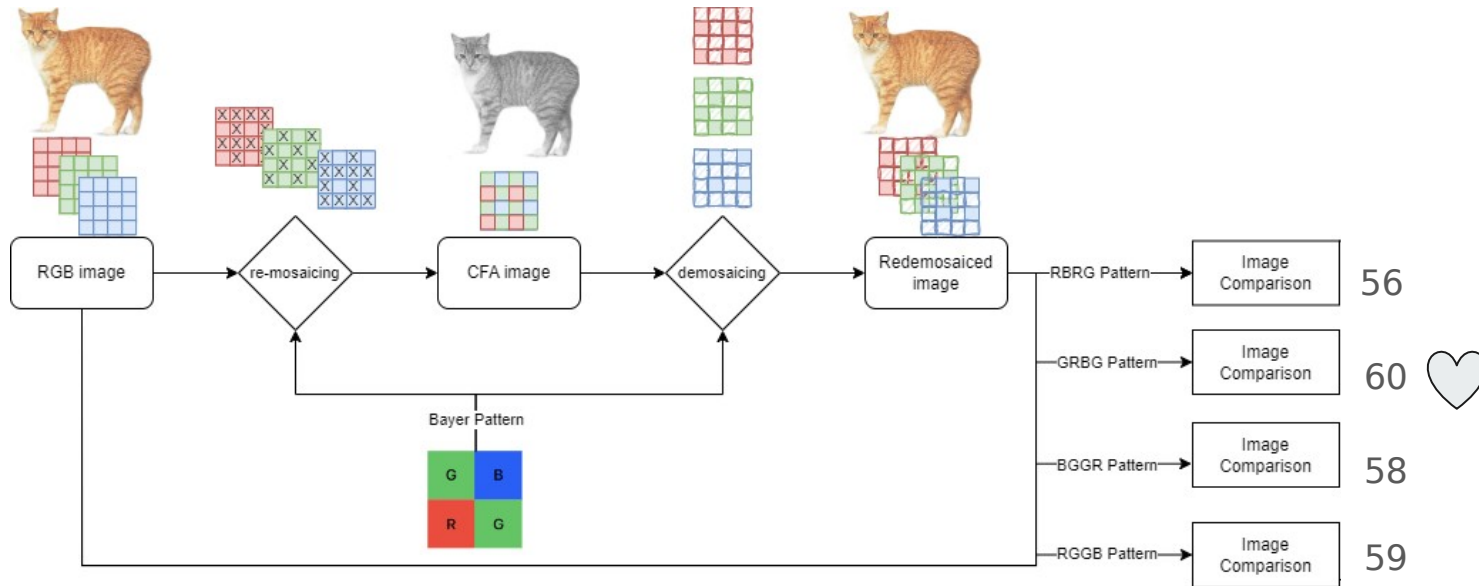


Quality 25 Compressed RAISE 1000 DeltaE2000, mean=0.0019, min=0.0001, max=0.0184
 RAISE 1000 DeltaE2000, mean=0.0877, min=0.0013, max=0.6525
 Quality 75 Compressed RAISE 1000 DeltaE2000, mean=0.0067, min=0.0002, max=0.4046





Bayer Pattern Prediction Accuracy



Bayer Pattern Prediction Accuracy

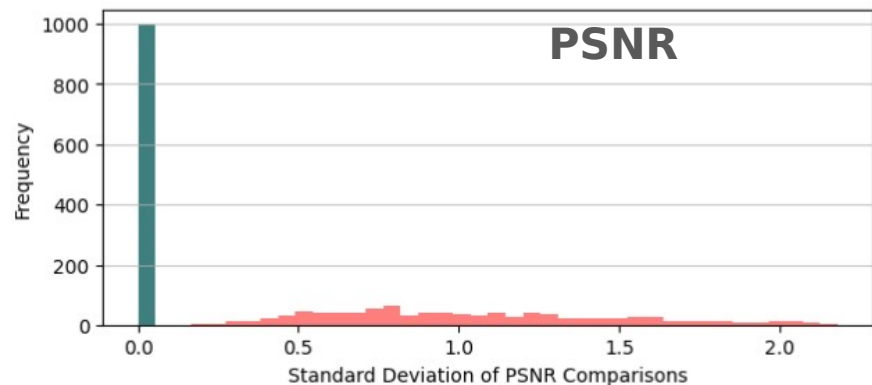
	PSNR	SSIM	VMAF	DeltaE
Uncompressed	58%	57%	61%	57%
25-Quality compressed	25%	17%	72%	18%
75-Quality compressed	43%	34%	71%	41%



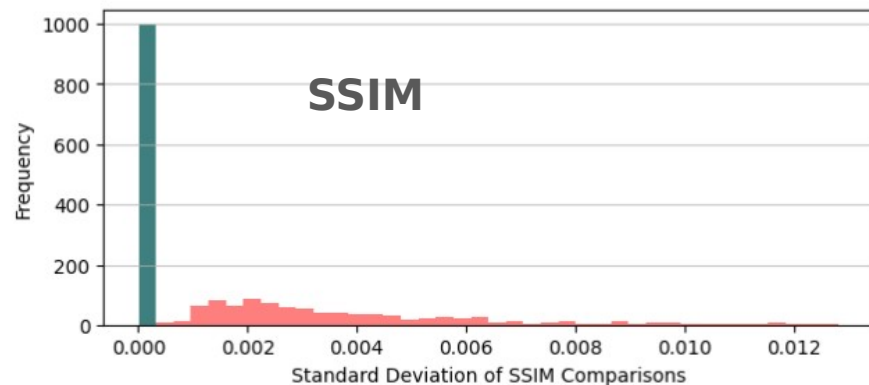
Resized Real Images as Reference

- Instead of using original images in the RAISE 1000 dataset
- Use resized images
 - Resize each dimension by a factor of 0.25
 - Resize each dimension by a factor of 0.5

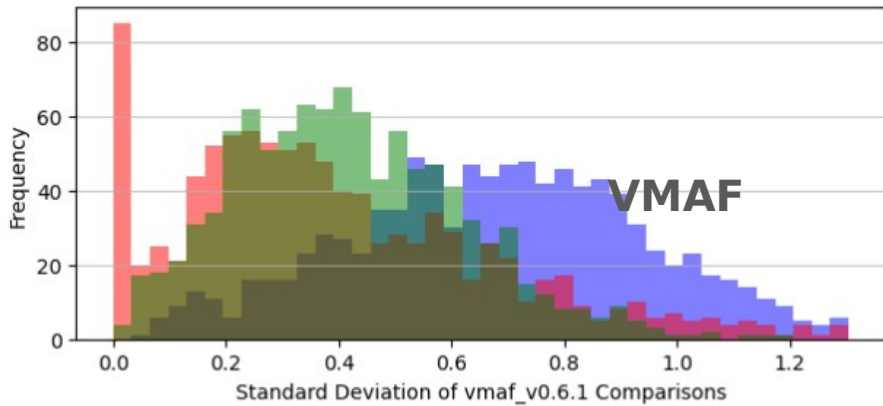
RAISE 1000 Resized by 0.25 PSNR, mean=0.0102, min=0.0001, max=0.0572
 RAISE 1000 PSNR, mean=1.1279, min=0.0263, max=3.1257
 RAISE 1000 Resized by 0.5 PSNR, mean=0.0057, min=0.0003, max=0.0394



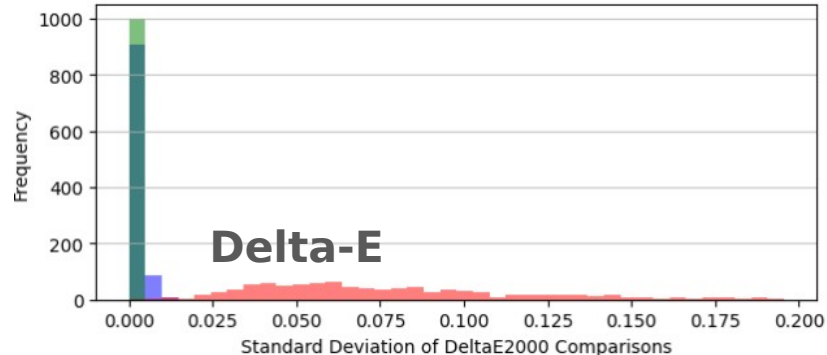
RAISE 1000 Resized by 0.25 SSIM, mean=0.0001, min=0.0000, max=0.0004
 RAISE 1000 SSIM, mean=0.0046, min=0.0001, max=0.0252
 RAISE 1000 Resized by 0.5 SSIM, mean=0.0000, min=0.0000, max=0.0001



RAISE 1000 Resized by 0.25 vmaf_v0.6.1, mean=0.7043, min=0.0590, max=3.3746
 RAISE 1000 vmaf_v0.6.1, mean=0.4304, min=0.0000, max=4.9799
 RAISE 1000 Resized by 0.5 vmaf_v0.6.1, mean=0.4209, min=0.0000, max=1.7704



RAISE 1000 Resized by 0.25 DeltaE2000, mean=0.0024, min=0.0001, max=0.0127
 RAISE 1000 DeltaE2000, mean=0.0877, min=0.0013, max=0.6525
 RAISE 1000 Resized by 0.5 DeltaE2000, mean=0.0005, min=0.0000, max=0.0026



Bayer Pattern Prediction Accuracy

	PSNR	SSIM	VMAF	DeltaE
Unresized	58%	57%	61%	57%
x0.25	26%	27%	72%	26%
x0.5	26%	25%	72%	23%

Compressed/Resized Real Images

- **Shift the** standard deviation **distributions** similar to fake images.
- The approach **will not work on altered real images.**

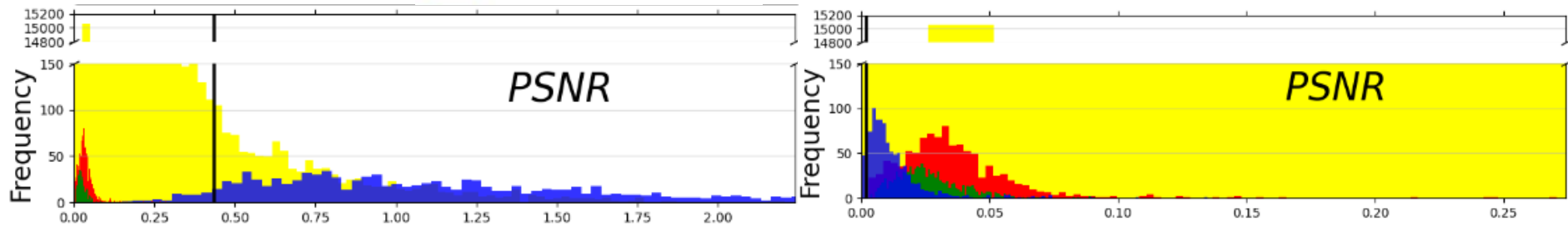


Other Demosaicing Algorithms

- Realistically we **do not know** what is the proprietary **demosaicing algorithms used by the camera manufacturers**.
- The **performance may depend** on the choice of **demosaicing algorithm**.
- Tried other demosaicing algorithms and compare the results.
 - Edge-aware demosaicing
 - Variable number of gradients demosaicing
 - Bilinear demosaicing

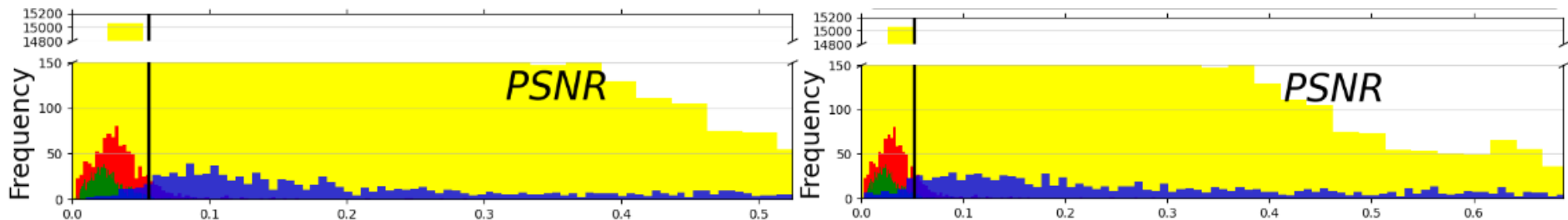
Distributions of Image Comparison Standard Deviations

RAISE 1000 DiffusionDB 1000
GenImage Landscape PhotoReal



Malvar et al. demosaicing

Variable number of gradients demosaicing



Edge aware demosaicing

Bilinear demosaicing



Other Demosaicing Algorithms

- In general, **Malvar et al. demosaicing** works really well.
- Unsure if the observation generalize to other cameras.

Summary

- A novel approach to differentiate fake images from real images
 - Works as long as the Bayer patterns in the real images are not broken
- Future works
 - Explore how much do compression and resizing shift the distributions
 - Repeat the experiment on other real images taken by cameras from other manufacturers.

